

Advanced Automation Control in an Ambulance under Emergency Condition

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Abstract—Currently, India is facing huge traffic congestion and the traffic disturbance in many major cities around is very severe. Mainly in urban areas, most of the people are using cars as transport when they go out. Due to this traffic blockage, there is a rise in road accidents which direction to a ruin of individual human lives. To avoid this we have implemented a scheme which can control the traffic signals automatically in its path way and reduce the amount of traffic congestion at the signal. The ambulance is implemented with embedded system units, which finds the accident spot and delivers the spot to the close by ambulance through GPS. The ambulance guides the traffic lights in the roadway to the hospital and it also checks the next consecutive signal to shorten the time loss. The vehicle unit is equipped with vibration sensor to determine the vibration if it exceeds the level. The embedded unit sense the accident happens and its sends the neighborhood ambulance unit through wireless transmission. RFID reader is needed to extricate the ambulance from other vehicles.

Keywords— *Vibration sensor, Global system for mobile communications(GSM), Global positioning system(GPS) , micro controller, Radio Frequency Identification Reader (RFID).*

I. INTRODUCTION

However human life is in great threatening because of this hike in traffic jam. The soul of victim is also in emergency when there is a lagging in conveying the notification to the ambulance immediately. There must be some pre programmed unit finds the accident spot and sends the announcement to the control unit and determine the accident spot, so that ambulance grasp in time and moves to the hospital. Programmed brake systems are used nowadays, but they only track front crash but they won't track rear end crash. It depends upon the driver alertness and his skill

Furthermore these advanced intelligent techniques have complex programs to determine accident insecurity. Anyhow these algorithms can't act as a driver's thoughts in simulation under slack and braking [2]

In this paper we have concentrated primarily on finding the accident spots and supervising the traffic lights in the roadway to decrease the time lagging of the ambulance in going up to the hospital. The notification from the GPS the ambulance is tracked to the accident vicinity region[1]. The RF transmitter is reserved and it's traffic signal is switch to green for the ambulance to rush through without lagging and thus by using these data the ambulance is instructed to reach the hospital by the embedded unit[3]. The authority wise traffic signal changing happened when two ambulances reach at the traffic signals at the same instantaneous of time, the one which comes first will be send first. The RF transmitter the data is processed to the traffic signals and it is supervised [1].

II. RELATED WORK

In the reference [4], the automatic traffic light switching has been characterized. To make sure, that the ambulance reach the accident location and provide better treatment to the victim

An adjuster switch is used to shift the traffic lights to green at any time the ambulance reaches the adjacent signal. We can reduce the held up of the ambulance in reaching the hospital by altering the traffic lights to green. The signals in the pathway in which the ambulance is instructed are accordingly shift one at a time when the ambulance goes near to it. The instruction to the ambulance section and the traffic signal sections are passed through RFID transmitter.

III. BLOCK DIAGRAM OF PROPOSED SYSTEM

Advanced programmable control the traffic lights are turned significantly from the ambulance unit. The system consists of vehicle section, ambulance section, traffic signal section.

A. Vehicle Unit

Vehicle section all the components are guarded by the embedded unit. Vibration sensor is used to find the changes in reverberation and will send this notification to the embedded unit by change into electrical charge [1]. The

micro controller correlate this value with the stable frequency value which is been set in the vehicle and if it equivalent or excel the limits, then the GPS finds the exact position of the vehicle and gives this information to the embedded controller unit. The microcontroller sends this notification to the vehicle unit by using zigbee and max232 where max232 is purposely used for voltage adjustment. The vehicle unit is shown in the below figure1. GPS trace out the accident location by using latitudes and longitudes.

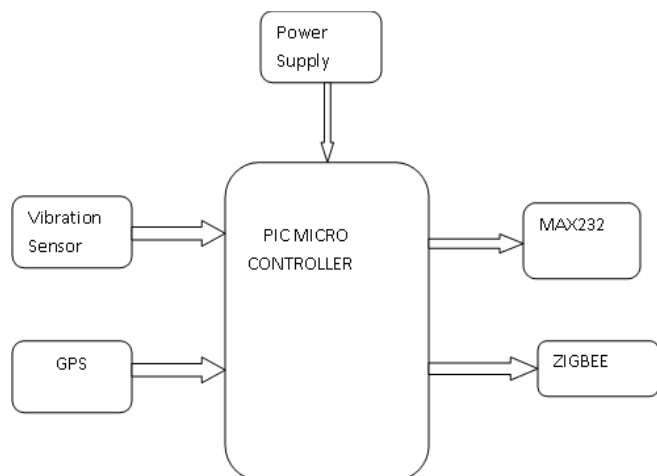


Fig. 1. Vehicle Unit

By using PIC controller as the micro controller unit, which is the main assistant for the full unit. Vibration sensor senses whether accident has occurred. The reading is shown on the lcd [2].Here zigbee is used to pass the notification midway between vehicle and ambulance.

B. Ambulance Unit

Intelligent unit is used in the ambulance unit. When the ambulance unit gathers the GPS location of the disaster region from the vehicle unit and thereby immediately the ambulance has ability to reach the nearest to the disaster region. The Embedded controller structure will communicate to ambulance unit is union with the RFID tag and this RFID tag will be effortless to control all the traffic lights in its path way to the hospital by turn out without any late to reach the hospital [4].Thus the RFID tag is used to communicate with the traffic light .The ambulance unit senses and change the switch to connect with the traffic unit. The ambulance unit not only controls the neighborhood traffic signal, but also the other successive traffic signals on its roadway to the hospitals with a time lagging of 15 to 20 seconds.

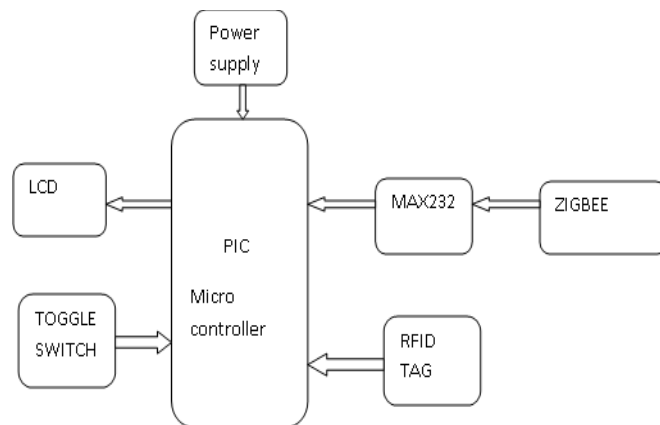


Fig. 2. Ambulance Unit

The ambulance section unit shown in figure2.The ambulance consisting of a RFID tag to find out the difference from the other general transport. The RFID reader is mounted in all the traffic signal it has to be controlled.

C. Traffic Signal Unit

The embedded pic controller in the traffic unit draws the data about the disaster region of a vehicle. The RF receiver in the section collects the information about the approaching special ambulance proximity to the traffic signal. This emergency ambulance closely proximity to the traffic signals the traffic signal lights will automatically toggle to green lights through the RFID tag. [4]. The certain ambulance is granted to reach the hospital immediately. The purpose of decoder, encoder and signal indicator for sending the data. The reader device collects the data from the ambulance section which the RF transmitter passes from the ambulance section. The traffic signal section shown in figure3.

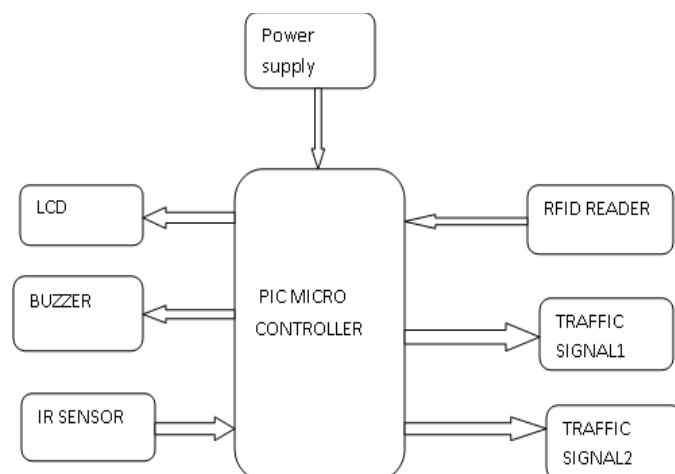


Fig. 3. Traffic Signal Unit

V. HARDWARE RESULTS

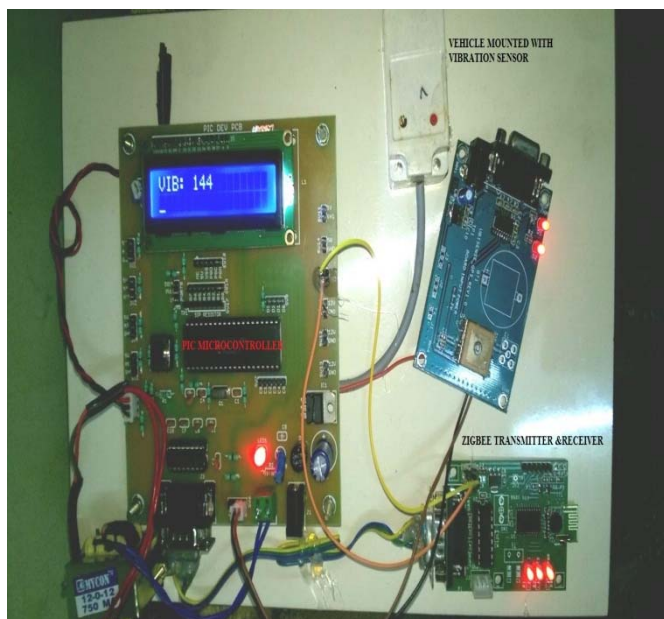


Fig. 7. Hardware setup for Vehicle section

In vehicle section whenever the crash occurs the vibration sensor sense the accident and the accident spot location is processed to the vicinity ambulance. The instruction is processed through zigbee. Max232 is used here to govern the increased flow of current. Whenever the limit has exceeded the setup frequency the LCD displays the accident spot taken place.

In vehicle and ambulance section there will be two max 232 .one is located on the embedded pic microcontroller IC and another will be located on the zigbee IC .Max232 are needed for the working purpose of the RS232 wire which will send the information from one unit to the other unit, but this RS232 will function only for the voltage amplitude between 5-12V and this voltage is supplied to the RS232 wire by the one.Max232 which is on the pic MC IC and then the RS232 works ,but if the same voltage continues to the zigbee IC then the zigbee will be destroyed because zigbees will be working only between the range 0-5V and thus this voltage is maintained in the zigbee with the help of 2.max232 which is in the zigbee IC and thus the max232 works by increasing the needed voltage to the RS232 wire and by decreasing the needed voltage to the zigbee IC

Fig 8. represents the hardware diagram of the ambulance section .It catch-up the information from the vehicle section through zigbee and arrives to the accident location spot by using latitudes and longitudes connected with GPS.



Fig. 8. Hardware setup for Ambulance section

The below picture represents the programmable control of traffic lights from red to green .At any time the special ambulance proximity to the traffic signal, the RFID tag reader collects the data and process the data to the pic microcontroller to toggle the lights to green .The data is processed the instruction to the forthcoming traffic signal with a stoppage of 10 to 15 seconds.

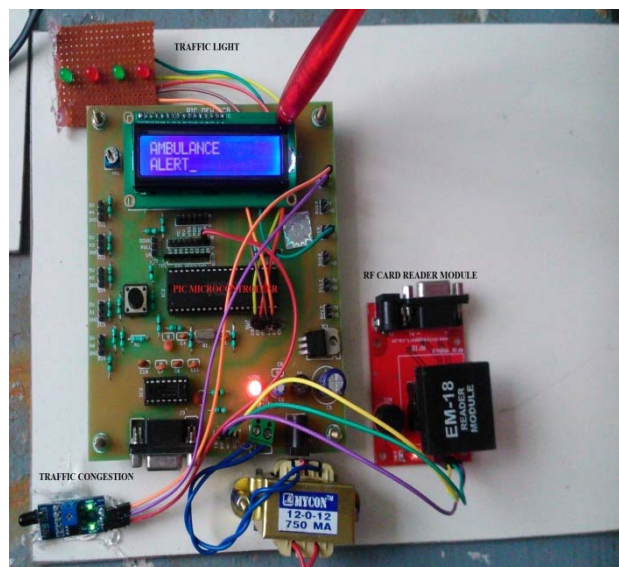


Fig. 9. Hardware Implementation for Traffic section

VI. CONCLUSION

In this paper, intelligent traffic light toggling is proposed for controlling the traffic signals endeavor of ambulances during the disaster accident condition. Thus if this system is accomplished in countries with huge residents like our nation, can outcome with better development. This system is more authentic with no loss of time delay saving more human life.

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